

Roll No.

TBT-504(b)

B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

STEM CELL TECHNOLOGY

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Differentiate between Embryonic Stem cells and Adult Stem cells.

(CO2, CO4)

OR

- (b) Highlight the applications of Stem cells in Gene therapy. (CO2, CO4)

2. (a) Diagrammatically classify the stem cells on the basis of potency and their properties.

(CO1, CO2)

OR

- (b) "Technique used to detect and analyze the chemical and physical characteristics of cells or particles." Comment on such technique and how it can identify stem cells.

(CO1, CO2)

P. T. O.

(2)

3. (a) Discuss the techniques involved in fate mapping of stem cells. Give example of one such species and determine its fate diagrammatically.

(CO1, CO2)

OR

- (b) Explain the isolation and culture process of stem cells with the help of a diagram.

(CO1, CO2)

4. (a) "Cells derived from skin or blood cells that have been reprogrammed back into an embryonic-like pluripotent state." Describe such stem cells and how it can aid in the disease treatment.

(CO2, CO4)

OR

- (b) What are mesenchymal stem cells ? Where are they found ? Give its therapeutic application in cell biology.

(CO2, CO4)

5. (a) Give the relevance of plasticity and cell differentiation in cancer cells.

(CO4, CO2)

OR

- (b) Discuss the activation process of neural stem cells.

(CO4, CO2)